FLATNESS AND FAITHFUL FLATNESS ARE PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tensor-hom adjunction as an identification of tensor products. Used at the first line of the counted proof (lines 9112-9114). Identifies N tensor over R' of R' tensor M with N tensor over R of M for every R'-module N.
- Flatness as exactness of tensoring with M. Used at the last line of the counted proof (lines 9114-9116). Transfers the exactness property from the functor tensoring with M over R to the base changed functor.

Lemma 0.1. *Suppose that M is (faithfully) flat over R, and that $R \to R'$ is a ring map. Then $M \otimes_R R'$ is (faithfully) flat over R'.*

Proof. For any R'-module N we have a canonical isomorphism $N \otimes_{R'} (R' \otimes_R M) = N \otimes_R M$. Hence the desired exactness properties of the functor $- \otimes_{R'} (R' \otimes_R M)$ follow from the corresponding exactness properties of the functor $- \otimes_R M$. □

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